



A Report on

One day workshop on “Robotics System MATLAB Toolbox”

by Mr. Kunal Khandelwal,

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Organised By

THE ROBOTICS SOCIETY STUDENT CHAPTER

23-09-2023



A ONE-DAY WORKSHOP ON ROBOTICS SYSTEM MATLAB TOOLBOX

About the Workshop:

The Robotics Society (TRS), the student chapter at Birla Vishvakarma Mahavidyalaya, an autonomous institution managed by Charutar Vidya Mandal and accredited with an 'A' grade by NAAC, is organizing a one-day Workshop on "Robotics System MATLAB Toolbox" on September 23, 2023, with objective of helping participants to gain a comprehensive understanding of key aspects of Robotic Systems, its implementations and applications.

Key Speakers:

Eminent speakers from MathWorks-DesignTech Systems Pvt. Ltd. will be conducting the sessions.

Topics Covered:

- 1.Introduction and Overview of Robotic Systems
- 2.Modelling Physical Systems (Robot Arm)
- 3.Modelling Electrical Systems (DC Motor)
- 4.Developing Control Systems (PID Controller)
- 5.Deploying to Hardware (Support Packages)

Registration Fee:

- Rs. 250/- for Non TRS members.
- No fees for TRS BVM student chapter No members.
- Fees include participation kit, tea, and lunch.

Who can attend:

Faculty members, UG/PG/PhD Students from any discipline of engineering can attend the workshop.

 Date: September 23, 2023
 Time: 10:00 am to 5:00 pm
 Venue: FSM Lab (H106) BVM

Register Here:



Limited seats available

Contact for Inquiries:

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B.V.M Engineering College [An Autonomous Institution]

Managed By Charutar Vidya Mandal

TRS students chapter organized

One day workshop on Robotics System MATLAB Toolbox”

Event Overview:

The Robotics Society (TRS) at Birla Vishvakarma Mahavidyalaya proudly conducted a one-day workshop on September 23, 2023, focused on the "Robotics System MATLAB Toolbox." This event aimed to empower participants with a comprehensive understanding of fundamental aspects of robotic systems, offering insights into various facets of robotics technology and hands-on experience.

Inaugural Ceremony:

The Robotics Society Student Chapter organized the one day workshop on "Robotics System MATLAB Toolbox” on September 23, 2023 from 10:00 am to 5:30 pm. It was inaugurated by Dr.Indrajit Patel, Principal BVM, Dr. Vinay Patel, Head Mechanical Dept and Advisor TRS student chapter, Mr. Kunal Khandelwal, An Application engineer –Mathworks, Design Tech Systems Pvt Ltd. Dr. Tanmay Pawar, Head Electronics Department, Dr. Bhargav Goradia, Head EC dept, Dr. Mehfuza Holia and Prof. Ashish Thakkar , Event coordinator. Total 50 students enthusiastic students had the unique opportunity to engage in hands-on activities, thereby embarking on a journey of learning and innovation

Objectives:

1. To offer participants a comprehensive grasp of the fundamental components of Robotic Systems, encompassing introductory insights and overviews of Robotic Systems.
2. To facilitate the modeling of physical systems, with a specific focus on Robot Arm simulations, to deepen participants' understanding of how robotic mechanisms operate.
3. To provide participants with an understanding of the electrical components of robotic systems, particularly the modeling of DC Motors, enabling them to appreciate the interplay between electronics and robotics.
4. To impart knowledge and practical experience in developing control systems, with a specific emphasis on PID Controllers, a critical element in regulating robotic actions effectively.
5. To familiarize participants with the process of deploying robotic systems to hardware using Support Packages, bridging the gap between theoretical knowledge and real-world application.
6. To emphasize the significance of hands-on experience as an integral and transformative aspect of the workshop, allowing participants to apply theoretical concepts in a practical context, thereby enhancing their learning experience."

Key Speaker:

The sessions were conducted by Dr. Kunal Khandelval, representing MathWorks-DesignTech Systems Pvt. Ltd. Dr. Khandelval's extensive expertise in the field and his association with MathWorks guaranteed that participants benefited from top-notch instruction regarding the Robotics System MATLAB Toolbox. Workshop Highlights

Outcomes:

1. In-Depth Understanding: Dr. Kunal Khandelval's sessions proved to be highly instructive, providing participants with a profound understanding of the Robotics System MATLAB Toolbox. Attendees gained comprehensive insights into the intricacies of Robotic Systems, spanning from foundational concepts to practical real-world applications.

2. Practical Application of Knowledge: The hands-on segment of the workshop empowered participants to put their newly acquired knowledge into practice, fostering a practical and tangible understanding of the subject matter. This interactive approach ensured that attendees left the workshop not only well-informed but also equipped with the skills to effectively utilize the MATLAB Toolbox within the realm of robotics.

3. High Attendance and Enthusiastic Participation: The workshop garnered significant interest and was well-attended, attracting a diverse audience including students, faculty members, and robotics enthusiasts. The enthusiastic participation, coupled with the engaging sessions and expert guidance, contributed to the overall success of the event, reinforcing its value as a knowledge-sharing platform.

Glimpses of an Events:

